

## Vortex Ring (Smoke Ring)

## Pause

Reset

Cross-section of a vortex ring: counter-rotating cores translate themselves at  $V_{\text{self}} \approx \Gamma / (4\pi a)$   
 $V_{\text{self}} \approx 0.2 \text{ px/s}$



## VORTEX RING

Circulation  $\Gamma$  185

Half-separation a 66 px

Core radius  $r_c$  28 px

## DISPLAY

☒ Show vortex cores

Side view (cross-section) of a thin axisymmetric vortex ring shown as a pair of counter-rotating Lamb–Oseen vortices at  $y = \pm a$ . Smoke particles are passively advected by the analytic velocity field; the ring drifts horizontally at the textbook self-induced speed  $V_{\text{self}} = \Gamma / (4\pi a)$  (no viscosity, no compressibility).

## MEASURED VALUES

185

a 66 px

|  |        |           |
|--|--------|-----------|
|  | V_self | 0.22 px/s |
|--|--------|-----------|

## About this model

A smoke ring is shown in cross-section as a pair of counter-rotating Lamb–Oseen vortices of strength  $\pm\Gamma$  separated by  $2a$ . Each core induces a velocity field  $u_\theta(r) = (\Gamma/2\pi r) \cdot (1 - \exp(-r^2/r_c^2))$  that is regularised at the centre by the viscous core radius  $r_c$ . By symmetry, the two vortices push each other forward at the self-induced speed  $V_{\text{self}} \approx \Gamma/(4\pi a)$  — exactly the mechanism that lets a real toroidal vortex propagate through still air. Tracer particles seeded around the cores are advected by the combined velocity field and reveal the rolled-up smoke pattern.

**Who it's for:** Intro fluid dynamics and aerodynamics; complements the airfoil and Kármán-vortex labs.

### KEY TERMS

- vortex ring
- Lamb–Oseen vortex
- self-induced velocity
- circulation  $\Gamma$
- toroidal vortex
- inviscid limit

### How it works

A vortex ring is two **counter-rotating** vortex tubes joined into a torus. In a 2D cross-section it looks like a vortex pair that **propels itself** through the surrounding fluid at  $V_{\text{self}} \approx \Gamma / (4\pi a)$ .

### Key equations

$$V_{\text{self}} \approx \Gamma / (4\pi a) \text{ (thin ring, inviscid)}$$

$$v_\theta(r) = (\Gamma / 2\pi r) \cdot (1 - \exp(-r^2/r_c^2)) \text{ (Lamb-Oseen core)}$$

## Frequently asked questions

### Why do the two vortices move forward instead of just rotating?

Each vortex sits in the velocity field induced by the other one. Because their circulations have opposite sign, the field they impose on each other points the same way — forward — so the pair drifts together at  $V_{\text{self}} \approx \Gamma/(4\pi a)$ .

### What is the role of the core radius $r_c$ ?

A pure point vortex has infinite velocity at the centre. The Lamb–Oseen profile smoothes that singularity over a viscous core of size  $r_c$ , giving a finite peak speed and a smooth tracer field — closer to a real, slightly viscous ring.

Why a 2D cross-section instead of a real torus?

The defining dynamics — circulation, mutual induction, and self-propagation — are visible already in the slice that cuts the ring through its symmetry plane. Rendering the full 3D torus would hide the two-vortex structure that drives the motion.

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Ideal Line Vortex

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$v_\theta = \Gamma / (2\pi r)$  arrow field; circulation and potential flow cartoon.

NEW SCHOOL

Bernoulli Flow

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Pipe narrows: speed up, pressure down. Continuity + Bernoulli.

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Coriolis Effect

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Puck on a rotating platform: curved path in the rotating frame vs straight line inertial.

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Magnus Effect (Ball)

▶ Launch Simulator

Same  $v_0$  and  $\theta$  with vs without spin: toy  $a = (k\omega v_y, -g - k\omega v_x)$ ; range comparison.

NEW FEATURED UNIVERSITY / RESEARCH

Double Pendulum

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Mesmerizing chaotic motion with path tracing.

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Foucault Pendulum (Sketch)

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$\Omega_{\text{eff}} = \Omega_E \sin|\lambda|$  with time scale; top view rotating swing line.